

Square-root Exponential Counterexamples to the Affine Positivity Conjecture

Run `fa_banach_001`, lane 11

August 9, 2026

Status: candidate counterexample, likely valid. For every $p, \beta > 0$, a normalized multiple of $r^p e^{-\beta\sqrt{r}}$ has an everywhere nonnegative affine Wigner distribution but is not a generalized Klauder wavelet.

Source Conjecture

The source is Eirik Berge, Stine Marie Berge, and Franz Luef, *The Affine Wigner Distribution*, arXiv:1912.03112 [1]. The conjecture occurs in the subsection *Affine Positivity Conjecture* on printed pages 25–26 (PDF pages 25–26). Because the discussion crosses a page break, both source crops are included.

Motivated by Proposition 9.2, it is of interest to investigate the k -support constant C_k both numerically and asymptotically.

Affine Positivity Conjecture

One of the major results about the classical Wigner distribution is regarding positivity; when is W_f a non-negative function on $\mathbb{R}^{2d}$? Normalized functions $f \in L^2(\mathbb{R}^d)$ such that W_f is non-negative would generate probability density functions on $\mathbb{R}^{2d}$ that represent the time-frequency distribution of f . However, a well-known result of Hudson [15, Theorem 4.4.1] shows that this can only happen for suitably perturbed Gaussians.

Turning to the affine setting, we would like to determine the normalized functions $\psi \in L^2(\mathbb{R}_+)$ such that W_{Aff}^ψ is a non-negative function on the affine group. In [19] the authors showed that the affine Wigner distribution $W_{\text{Aff}}^{\psi_s}$ is non-negative if ψ_s is the so called *Morse ground state*

$$\psi_s(r) := \frac{r^s e^{-\frac{r}{2}}}{\Gamma(2s)}, \quad s \geq 0.$$

25

In our case, we will only consider ψ_s for $s > 0$ as $\psi_0 \notin L^2(\mathbb{R}_+)$. More generally, one can use the invariance properties (4.2) and (4.3) to show that the affine Wigner distribution W_{Aff}^ψ of

$$\psi(r) = C r^{-i(x+ia)} e^{i(y+ib)r}, \quad (9.2)$$

is non-negative when $C \in \mathbb{C}$ and $(x, a), (y, b) \in \text{Aff}$. The functions on the form (9.2) are the *generalized Klauder wavelets* in [9, Equation (41)] that are in $L^2(\mathbb{R}_+)$. This leads to the following affine positivity conjecture:

The only functions $\psi \in L^2(\mathbb{R}_+)$ such that W_{Aff}^ψ is non-negative on the affine group are the generalized Klauder wavelets in (9.2).

The exact concluding statement is:

The only functions $\psi \in L^2(\mathbb{R}_+)$ such that W_{Aff}^ψ is non-negative on the affine group are the generalized Klauder wavelets in (9.2).

Here the source uses $L^2(\mathbb{R}_+) = L^2(\mathbb{R}_+, dr/r)$, and its generalized Klauder wavelets are

$$\psi(r) = Cr^{-i(x+i\alpha)}e^{i(y+ib)r}, \quad C \in \mathbb{C}, \quad (x, \alpha), (y, b) \in \text{Aff}, \quad (1)$$

so $\alpha, b > 0$.

Definition and Claimed Counterexamples

Put

$$\lambda(u) = \frac{ue^u}{e^u - 1}, \quad \lambda(-u) = \frac{u}{e^u - 1}, \quad (2)$$

with the continuous value 1 at $u = 0$. The affine Wigner distribution used by the source is

$$W_{\text{Aff}}^\psi(x, a) = \int_{\mathbb{R}} \psi(a\lambda(u)) \overline{\psi(a\lambda(-u))} e^{-2\pi i x u} du, \quad x \in \mathbb{R}, \quad a > 0. \quad (3)$$

For $p, \beta > 0$, define

$$\psi_{p,\beta}(r) = N_{p,\beta} r^p e^{-\beta\sqrt{r}}, \quad N_{p,\beta} = \left(\frac{(2\beta)^{4p}}{2\Gamma(4p)} \right)^{1/2}. \quad (4)$$

The chosen constant makes $\|\psi_{p,\beta}\|_{L^2(dr/r)} = 1$.

Theorem 1. *For every $p, \beta > 0$, the function $\psi_{p,\beta}$ in (4) belongs to $L^2(\mathbb{R}_+, dr/r)$, and*

$$W_{\text{Aff}}^{\psi_{p,\beta}}(x, a) \geq 0 \quad (x \in \mathbb{R}, \quad a > 0).$$

It is not a generalized Klauder wavelet of the form (1). Consequently, the affine positivity conjecture as stated in arXiv:1912.03112 is false.

Proof Intuition

For each scale a , formula (3) is an ordinary Fourier transform in u . Thus it is enough to prove that its slice kernel is both integrable and positive definite. The geometric symmetrization built into the affine variables is especially well matched to the square root:

$$\sqrt{\lambda(u)} + \sqrt{\lambda(-u)} = \sqrt{u \coth(u/4)}.$$

After removing the value at $u = 0$, the kernel splits into two factors. The power factor is positive definite by Euler's product for $\sinh$. The square-root exponential is positive definite because $u \coth(u/4) - 4$ is conditionally negative definite and the classical square-root subordination formula expresses the desired factor as a positive mixture of Schoenberg kernels. This gives positivity for an entire two-parameter family rather than for a single numerical example.

Proof

We use two standard notions on the additive group $\mathbb{R}$. A continuous function F is *positive definite* if every matrix $[F(u_j - u_k)]_{j,k}$ is positive semidefinite. A real even function q with $q(0) = 0$ is *conditionally negative definite* if

$$\sum_{j,k} c_j \overline{c_k} q(u_j - u_k) \leq 0 \quad \text{whenever} \quad \sum_j c_j = 0.$$

We use Schoenberg's theorem: such a q is conditionally negative definite if and only if e^{-tq} is positive definite for every $t > 0$; see [2, Chapter 3].

Proof of Theorem 1. The norm computation is immediate after $t = \sqrt{r}$:

$$\int_0^\infty r^{2p} e^{-2\beta\sqrt{r}} \frac{dr}{r} = 2 \int_0^\infty t^{4p-1} e^{-2\beta t} dt = \frac{2\Gamma(4p)}{(2\beta)^{4p}}.$$

In fact, $x \mapsto \psi_{p,\beta}(e^x)$ is Schwartz on $\mathbb{R}$, so the example also lies in the source paper's class $\mathcal{S}(\mathbb{R}_+)$.

Fix $a > 0$ and suppress the harmless positive normalization constant. Set

$$h(u) = \frac{u}{2 \sinh(u/2)}, \quad h(0) = 1. \quad (5)$$

Then

$$\lambda(u) = h(u)e^{u/2}, \quad \lambda(-u) = h(u)e^{-u/2},$$

and hence

$$\lambda(u)\lambda(-u) = h(u)^2, \quad (6)$$

$$\sqrt{\lambda(u)} + \sqrt{\lambda(-u)} = 2\sqrt{h(u)} \cosh(u/4) = \sqrt{u \coth(u/4)}. \quad (7)$$

Define the nonnegative even function

$$f(u) = \sqrt{u \coth(u/4)} - 2, \quad f(0) = 0. \quad (8)$$

Dividing the slice kernel in (3) by its positive value at zero and using (6)–(7) gives

$$\Phi_a(u) := \frac{\psi_{p,\beta}(a\lambda(u))\psi_{p,\beta}(a\lambda(-u))}{|\psi_{p,\beta}(a)|^2} = h(u)^{2p} \exp(-\beta\sqrt{a} f(u)). \quad (9)$$

We first prove that h^{2p} is positive definite. Euler's product gives

$$\frac{\sinh(u/2)}{u/2} = \prod_{n=1}^\infty \left(1 + \frac{u^2}{4\pi^2 n^2}\right). \quad (10)$$

For every $c, q > 0$,

$$\left(1 + \frac{u^2}{c^2}\right)^{-q} = \frac{1}{\Gamma(q)} \int_0^\infty t^{q-1} e^{-t} e^{-tu^2/c^2} dt. \quad (11)$$

The functions e^{-su^2} are positive definite, so (11) is positive definite. Finite products of positive-definite functions are positive definite, and the pointwise limit in (10) is continuous at zero. Therefore

$$h(u)^{2p} = \prod_{n=1}^\infty \left(1 + \frac{u^2}{4\pi^2 n^2}\right)^{-2p} \quad (12)$$

is positive definite.

We next prove that the exponential factor in (9) is positive definite. The partial-fraction expansion of $\coth$ is

$$v \coth(v/2) - 2 = 4v^2 \sum_{n=1}^{\infty} \frac{1}{v^2 + 4\pi^2 n^2}. \quad (13)$$

For $c > 0$, the function $c^2/(v^2 + c^2)$ is positive definite (it is, up to a positive Fourier normalization, the transform of $e^{-c|x|}$). Since its value at zero is one,

$$\frac{v^2}{v^2 + c^2} = 1 - \frac{c^2}{v^2 + c^2}$$

is conditionally negative definite. The locally uniform positive sum in (13) is therefore conditionally negative definite. Rescaling and multiplying by a positive constant show that

$$q(u) := 2((u/2) \coth(u/4) - 2) = u \coth(u/4) - 4 \quad (14)$$

is conditionally negative definite and nonnegative.

Now $f(u) = \sqrt{4 + q(u)} - 2$. For $c > 0$, the square-root subordination identity

$$e^{-c\sqrt{z}} = \frac{c}{2\sqrt{\pi}} \int_0^{\infty} t^{-3/2} \exp\left(-\frac{c^2}{4t}\right) e^{-zt} dt \quad (z > 0) \quad (15)$$

and (14) yield

$$e^{-cf(u)} = \frac{ce^{2c}}{2\sqrt{\pi}} \int_0^{\infty} t^{-3/2} \exp\left(-\frac{c^2}{4t} - 4t\right) e^{-tq(u)} dt. \quad (16)$$

By Schoenberg's theorem each $e^{-tq(u)}$ is positive definite. The scalar weight in (16) is positive and has total mass one (evaluate at $u = 0$). Thus e^{-cf} is positive definite for every $c > 0$. Taking $c = \beta\sqrt{a}$, both factors in (9) are positive definite, so Φ_a is positive definite.

Moreover, $0 < e^{-\beta\sqrt{a}f(u)} \leq 1$, while

$$h(u)^{2p} = O(|u|^{2p} e^{-p|u|}) \quad (|u| \rightarrow \infty).$$

Hence $\Phi_a \in L^1(\mathbb{R})$. The unnormalized slice kernel in (3) is the positive multiple $|\psi_{p,\beta}(a)|^2 \Phi_a$, so it too is integrable and positive definite. Bochner's theorem followed by Fourier uniqueness says that the Fourier transform of a continuous integrable positive-definite function is a continuous nonnegative function. Formula (3) is precisely that Fourier transform. Therefore

$$W_{\text{Aff}}^{\psi_{p,\beta}}(x, a) \geq 0$$

for every $x \in \mathbb{R}$ and $a > 0$.

Finally, the modulus of a nonzero generalized Klauder wavelet (1) is

$$|C| r^\alpha e^{-br}, \quad \alpha, b > 0,$$

whereas the modulus of (4) is $N_{p,\beta} r^p e^{-\beta\sqrt{r}}$. Equality for all $r > 0$ would, after logarithmic differentiation and multiplication by r , force

$$p - \frac{\beta}{2}\sqrt{r} = \alpha - br \quad (r > 0),$$

which is impossible because $\sqrt{r}$ and r have different growth. Thus the examples are not generalized Klauder wavelets. $\square$

Verification Notes

The accompanying script `code/check_counterexample.py` independently checks the algebraic identities, conditional-negative-definiteness matrices, positive-definiteness Gram matrices, and FFT approximations to all affine Wigner slices for 60 triples

$$p \in \{0.1, 0.3, 1, 2.5\}, \quad \beta \in \{0.2, 1, 3\}, \quad a \in \{0.05, 0.3, 1, 5, 20\}.$$

Run it from the repository root with

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/counterexamples/\
  1912.03112_affine_wigner_sqrt_exponential_positive/\
  code/check_counterexample.py
```

The finite tests are diagnostics only. Positivity for all parameters and all phase-space points follows from the positive-definiteness proof above.

Novelty and Limitations

A bounded check on August 9, 2026 searched the run’s registry, solution, attempt, and proof-gap indexes and used arXiv-focused web queries for the paper/arXiv id, the exact phrase *Affine Positivity Conjecture*, combinations of *affine Wigner*, *nonnegative*, *generalized Klauder*, and *counterexample*, and the square-root-exponential family. It found the source paper and older background on positive Morse/Klauder states, but no later explicit resolution and no prior appearance of this counterexample family. The search was bounded and cannot establish priority. The claimed mathematical statement is exactly a disproof of the source conjecture; it does not attempt a replacement classification of all nonnegative affine Wigner states.

Human Review Recommendation

Send to expert review as a full counterexample. The most useful adversarial checks are: (i) the constants in (13) and (14); (ii) the square-root subordination formula (15); and (iii) the use of Bochner plus L^1 -integrability to upgrade the representing positive measure to the pointwise nonnegative Fourier transform.

References

- [1] E. Berge, S. M. Berge, and F. Luef, *The Affine Wigner Distribution*, arXiv:1912.03112, 2019.
- [2] C. Berg, J. P. R. Christensen, and P. Ressel, *Harmonic Analysis on Semigroups: Theory of Positive Definite and Related Functions*, Graduate Texts in Mathematics 100, Springer, 1984.